Angular 16 (Webpack-based) and Angular 19 (Vite + esbuild)

⚙️ Angular 16 — Webpack-based (using @angular-devkit/build-angular)

```
{
 "$schema": "./node_modules/@angular/cli/lib/config/schema.json",
 "version": 1,
 "projects": {
  "my-app": {
   "projectType": "application",
   "root": "",
   "sourceRoot": "src",
   "architect": {
    "build": {
     "builder": "@angular-devkit/build-angular:browser",
     "options": {
      "outputPath": "dist/my-app/browser",
      "index": "src/index.html",
      "main": "src/main.ts",
      "polyfills": ["zone.js"],
      "tsConfig": "tsconfig.app.json",
      "assets": ["src/favicon.ico", "src/assets"],
      "styles": ["src/styles.css"],
      "scripts": []
     },
     "configurations": {
      "production": {
       "budgets": [{ "type": "initial", "maximumWarning": "2MB" }],
       "outputHashing": "all",
       "optimization": true,
       "sourceMap": false,
       "extractLicenses": true
      },
      "development": {
       "buildOptimizer": false,
       "optimization": false,
       "vendorChunk": true,
```

```
          "extractLicenses": false,
          "sourceMap": true
        }
      }
    },
    "serve": {
      "builder": "@angular-devkit/build-angular:dev-server",
      "options": {
        "browserTarget": "my-app:build"
      },
      "configurations": {
        "production": {
          "browserTarget": "my-app:build:production"
        },
        "development": {
          "browserTarget": "my-app:build:development"
        }
      }
    },
    "server": {
      "builder": "@angular-devkit/build-angular:server",
      "options": {
        "outputPath": "dist/my-app/server",
        "main": "src/main.server.ts",
        "tsConfig": "tsconfig.server.json"
      }
    },
    "prerender": {
      "builder": "@angular-devkit/build-angular:prerender",
      "options": {
        "routes": ["/"]
      }
    }
  }
 }
}
```

🧩 Summary (Angular 16)

- Webpack-based pipeline
- Multiple builders (browser, server, dev-server, prerender)
- Complex config, lots of duplication
- SSR & CSR handled separately

⚡ Angular 19 — esbuild + Vite-based (using @angular/build)

```
{
 "$schema": "./node_modules/@angular/cli/lib/config/schema.json",
 "version": 1,
 "projects": {
  "my-app": {
   "projectType": "application",
   "sourceRoot": "src",
   "architect": {
    "build": {
     "builder": "@angular/build:application",
     "options": {
      "outputPath": "dist/my-app",
      "index": "src/index.html",
      "main": "src/main.ts",
      "tsConfig": "tsconfig.app.json",
      "assets": ["src/favicon.ico", "src/assets"],
      "styles": ["src/styles.css"]
     },
     "configurations": {
      "production": {
       "optimization": true,
       "sourceMap": false
      },
      "development": {
       "optimization": false,
       "sourceMap": true
      }
     }
    },
```

```
      "serve": {
        "builder": "@angular/build:dev-server",
        "options": {
          "buildTarget": "my-app:build"
        }
      },
      "ssr": {
        "builder": "@angular/build:ssr",
        "options": {
          "outputMode": "server",
          "server": "src/main.server.ts",
          "routesFile": "src/app/app.routes.server.ts",
          "renderMode": "hybrid"
        }
      }
    }
  }
}
}
```

🧠 Summary (Angular 19)

- Uses **@angular/build** (replaces @angular-devkit/build-angular)
- Internally powered by **Vite + esbuild**
- **Unified** SSR + CSR + Prerender pipeline under one builder
- **Simple, fast, predictable**
- Supports **Hybrid Rendering** (RenderMode.Client, RenderMode.Server, RenderMode.Hybrid)
- Much fewer configuration lines

Feature	Angular 16	Angular 19
Builder package	@angular-devkit/build-angular	@angular/build
Bundler	Webpack 5	esbuild + Vite
Dev server	Webpack Dev Server	Vite Dev Server
SSR support	Separate (server + prerender)	Unified via ssr target

Render modes	None (manual SSR)	RenderMode API (Hybrid Rendering)
Speed	Moderate (Webpack rebuilds)	⚡ Super fast (esbuild incremental)
Config complexity	Verbose & repetitive	Minimal & declarative
HMR (Hot Reload)	Partial	Instant via Vite

🚀 What This Means for You

Migrating from Angular 16 → 19:

- Replace builders in angular.json with the new @angular/build equivalents.
- Remove unnecessary sections like polyfills, vendorChunk, etc.
- Update SSR setup to use "outputMode": "server" and RenderMode values.
- Angular CLI automatically detects modern build pipeline (esbuild + Vite).